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1 Open and closed sets

Definition. An interval on the number line is called open if it goes from A to B without containing A and B, ie it does not contain its boundary.

The same can be said for subsets of the plane or hyperplanes – they are open if they do not contain any of their boundary. They are closed if they contain all of their boundary. The formal definition of a boundary is a point where no matter how small of a circle or ball you make around that point, it will contain points both inside and outside the set. Note that since an open set contains none of its boundary, it has the property that for any point in the open set, all points sufficiently close to that point are in that set, so open sets must always have some thickness to them.

This is related to the idea that there is no smallest real number greater than another real number, and whether sets are open or closed turns out to be surprisingly important. Also, arbitrary unions of open sets are open since each point has a “ball” around it inside each of the open sets in the union and thus the whole thing, similarly arbitrary intersections of closed sets are closed. The reverse implications only hold for finite unions and intersections (since infinitely many balls intersection could be a point but finitely many is a ball). Sets can be both open and closed, and sets can be neither open nor closed.

A useful way to think of closed sets as sets where the set of everything not in the closed set forms an open set.

We usually call open intervals (a,b) meaning everything from a to b not inclusive, and closed intervals $[a,b]$ to mean everything from a to b inclusive, and $[a,b)$ means everything from a to b including a but not b.

2 Properties of the lebesgue integral

2.1 Preliminary lemmas

We will have to build a lot of boring theory and definitions that may seem irrelevant to get to the results we need. While this may look intimidating, I will aim to make this more self contained than all of the other proofs I have seen of results like this, which annoyingly are all hard to follow because they are aimed at people who are better at maths than me even though (at the time of first writing this) I am at the level where I am sometimes expected to implicitly use these results, something I aim to fix in this document for future interested students like me. No more “Let (Symbol I’ve never seen) be a measure space (steps with symbols I haven’t seen that I can’t follow) intimidation” nonsense.

Theorems about Lebesgue integrals here for our purposes pre-suppose that limits and integrals in question are well defined before proving they are equal to anything.

I mentioned in Level 4 that we will not fully define “Length”. However, I will explain it briefly here:

Definition. If we have a finite collection of disjoint open or closed intervals, it is clear what its total length should be. If we have sets $S_1 \subseteq S_2 \subseteq \dots$ that go over all the positive integers where we have already defined the length for each of them, then the length of the union of all S_i 's is equal to the limit of their lengths. We can assume for our purposes that measurable sets are those that can be made in this way. If $S \subseteq T$ and T has length l and S has length m then the set of everything in T but NOT in S has length $l - m$. Adding arbitrary subsets of measure 0 sets gives the Lebesgue measure. This is NOT to be confused with when I mention “measure 0” in any level 8 context since I don't actually prove that this definition and that definition are equivalent until later than that.

We can check that this is well defined. Here is how we would do that: If two choices of sequences of sets with known length that were strictly growing as above (Say S_n and T_n) had the same union and got different results then consider the intersections $S_n \cap T_m$, then this always has less length than T_m so as we increase m this will be in the limit S_n and have at most the length of the union of T_m , and this is true for all n so the length of the union of S_n is $\leq$ the length of the union of T_m and similarly the other way around so they are equal.

From now on, I will use intuitive properties of length instead of nailing down every technical detail.

We need to recall the definition of the Lebesgue integral from the end of level 4. Now we prove a useful lemma known as the monotone convergence theorem.

Lemma. Suppose we have a sequence of non-negative functions $0 < f_1 < f_2 < f_3 \dots$ converging to f in the limit pointwise (ie, f is the pointwise supremum), and suppose all f_n 's and f are integrable (ie its integral is finite) and all sets mentioned are measurable. Then $\int f_n \rightarrow \int f$. It is also true and we will see in the proof that the Lebesgue integral of f exists if we merely assume it exists for all f_n 's, but for riemann integrals you need the added hypothesis that f is integrable.

Proof. To do this, we will prove that eventually, $\int f_n$ eventually gets larger than $(\int f) - \varepsilon$ regardless of how small we make ε .

We know we can find a simple non-negative function $g(x)$ such that $\int g \geq (\int f) - \frac{\varepsilon}{3}$ because $\int f$ is defined as the least upper bound of integrals of simple functions, meaning if we could not find simple functions below f whose integrals are arbitrarily close to the integral of f , f would not be the least upper bound. The reason for our choice of $\frac{\varepsilon}{3}$ will become clear eventually. Now we want to shift our g downwards by a constant δ in such a way that $\int g^- \geq (\int f) - \frac{2\varepsilon}{3}$ where g^- is defined as $g - \delta$ if $g > \delta$ and 0 otherwise. This ensures g^- is always non-negative and that it is at most $g - \delta$. So we have two cases: If the length of our interval that we are integrating along is a finite length l , then we pick δ to be $\frac{\varepsilon}{3l}$, since then the total amount that the rectangles get moved down by which is $\frac{\varepsilon}{3l}$ times the total length of the rectangles which is l is at most $\frac{\varepsilon}{3}$. If we are integrating from $-\infty$ to ∞ then we can still apply the same argument since the length on which our g is non-zero is finite. This is true because the integral is defined as the least upper bound of the integrals of all such g .

Now define the set S_n as the set of values x such that $f_n(x) \geq g^-$. We will discuss why this is measurable shortly. Then since the f 's are increasing, each set S_n must contain the previous set S_{n-1} .

Now I claim there is an N such that the integral over the parts of our interval not containing S_N of f is less than or equal to $\frac{\varepsilon}{3}$. To do this, suppose the contrary, that there is a set T such that the integral of f over T is more than $\frac{\varepsilon}{3}$ but no point in T ever goes into any of the S_n 's. We know that f is not 0 anywhere in T because if $f(a)$ is 0 then a is in all S_n 's since all f_n 's will be 0 at those points by definition which means they satisfy $f_n(x) \geq g^-$ since g^- would also be 0 at those points as g^- is non negative and never bigger than f which is 0.

But then, this means that everywhere in T , g^- is strictly less than f , since if $0 < f < \delta$ then $g^- = 0$ and otherwise $g^- = f - \delta$. Therefore, $f_n(x) \leq g^-$ for all n and x in T by definition of the S_n 's and of T , so the least upper bound of $f_n(a)$ for any point a in T is at most $g^-(a)$ which is strictly less than $f(a)$, which contradicts the definition of f . This means if we let T_N be the interval not including S_N , then $\int_{T_N} f \leq \frac{\varepsilon}{3}$. Therefore, putting everything together,

$$\int f_n \geq \int_{S_N} f_n \geq \int_{S_N} g^- = \int g^- - \int_{T_N} g^- \geq \int g^- - \int_{T_N} f \geq \int g^- - \frac{\varepsilon}{3} \geq \left(\int f \right) - \varepsilon$$

Why the set mentioned above is measurable: If a function is, say, continuous, then the set of points it is $\leq$ something contains its limit points, hence is closed, hence is the complement of an open set, which is a union of open intervals (or

open balls in higher dimensions, and it is a union of those with only rational centers and radii, which we will see shortly why we can enumerate these into a sequence), hence measurable. Even if it is not continuous it will usually be intuitive that measurability is not an issue. If it is piecewise continuous then we just have a finite union of measurable sets which is measurable, and if the set of discontinuity points is finite it doesn't affect anything. Furthermore, if each f_n is measurable (which means that each of the sets $f_n > x$ are measurable sets) then the set $\{f > L\}$ is the set of all points such that some f_n is $> L$ which is the union over n of the sets $\{f_n > L\}$, which is measurable since it is a countable (we will see what this means shortly) union of measurable sets, thus f is measurable.

This completes the proof of the lemma. □

Note also that the same argument can be used to prove the above theorem (and thus the theorems below that depend on it) for integrals of functions over 2D, 3D, or higher dimensional areas. This will be important later. The Bolzano weierstrass theorem from level 4 (that every bounded sequence has a convergent subsequence) and its consequences can be used provided the set is not only bounded but also **closed**, since we want sequences to not only converge but have their limit be part of the set.

Now we will prove another lemma known as **Fatou's lemma**.

Lemma. If we have a sequence of non-negative functions $f_n(x)$, then for every x define $f(x) = \liminf_{n \rightarrow \infty} f_n(x)$, then on any interval we have $\int f \leq \liminf_{n \rightarrow \infty} \int f_n$.

Proof. For all fixed x , we define $g_n(x)$ as $\inf_{k \geq n} f_k(x)$. Then we know that for any fixed x , this is not decreasing as chopping more terms of the start cannot make the infimum lower. We also have the following by stuff we have discussed so far:

We have $\int f = \int \sup g_n = \sup \int g_n$ where the second equality is because g is increasing so we can apply the monotone convergence theorem. Since $\int g_n$ is an increasing sequence, the $\liminf$ of this sequence is going to equal the limit of the sequence, as the infima of the tails are the same as the terms themselves. But the limit of an increasing sequence is the same as its supremum, so we now have that $\int f = \sup \int g_n = \liminf_{n \rightarrow \infty} \int g_n$. Because g_n is by definition not greater than f_n for all n at any input value, we have that $\int f = \sup \int g_n = \liminf_{n \rightarrow \infty} \int g_n \leq \liminf_{n \rightarrow \infty} \int f_n$, completing the proof of Fatou's lemma. From how we defined f , we have that $\int \liminf_{n \rightarrow \infty} f_n \leq \liminf_{n \rightarrow \infty} \int f_n$. □

Remark. We can also get that if f_n is bounded above by a function g with a finite integral on our interval for all n , then applying Fatou's lemma to $g - f_n$ gives us the reverse fatou lemma: $\int \limsup_{n \rightarrow \infty} f_n \geq \limsup_{n \rightarrow \infty} \int f_n$. We will use this to prove the DCT.

2.2 The dominated convergence theorem

This theorem is completely overpowered for making the applied nonsense in A levels/university maths rigorous.

The **dominated convergence theorem** is as follows:

Theorem. Suppose we have a sequence of functions $f_n(x)$ which for all x converge to a function $f(x)$ as n goes to infinity, and there is a function $g(x)$ which has a finite integral on the interval we are working in, and that $|f_n(x)| \leq g(x)$ for all n and all x , then we have that $\int \lim_{n \rightarrow \infty} f_n = \lim_{n \rightarrow \infty} \int f_n$, essentially giving us a condition for when we can swap limits and integrals.

For our purposes, we assume f is Lebesgue integrable, and that f_n is Lebesgue integrable for every n . Under these hypotheses it works for Riemann integrals too but only if the limit and every function in the sequence is actually Riemann integrable - in most practical cases this will not be an issue.

Proof. Recall how at some places in level 4 and level 6.1 we talked about the triangle inequality which says that $|a +$

$b| \leq |a| + |b|$, we will use this now to say that

$$|f - f_n| \leq |f| + |-f_n| = |f| + |f_n| \leq 2g$$

Note that f is at most g at all places since it is the pointwise limit of functions that are at most g so it would not make sense for it to exceed g . We also have by definition that $\limsup_{n \rightarrow \infty} |f - f_n| = 0$. Also, in this next step, we will use the fact analogous to the triangle inequality which says that $|\int h(x)| \leq \int |h(x)|$ for any function $h(x)$. For real numbers this is the case because h is always between $-|h|$ and $|h|$, so the integral of h will always be between the integral of $-|h|$ and the integral of $|h|$, so the absolute value of that will be at most the integral of $|h|$. For complex numbers (which I am doing to show that we can generalize this beyond real integrals), we have $\int h = re^{i\theta}$ so $|\int h| = r$ and $\int he^{-i\theta} = r$. Since r is real, we have that $\text{Re}(\int he^{-i\theta}) = r$. Since the real part of the integral is the integral of the real part, we have

$$\int \text{Re}(he^{-i\theta}) = r \leq \int |he^{-i\theta}| = \int |h|$$

Since $|\int h| = r \leq \int |h|$ this completes the proof. Now back to dominated convergence:

$$\left| \int f - \int f_n \right| = \left| \int (f - f_n) \right| \leq \int |f - f_n|$$

Now, we use the reverse Fatou lemma:

$$\limsup_{n \rightarrow \infty} \int |f - f_n| \leq \int \limsup_{n \rightarrow \infty} |f - f_n| = \int 0 = 0$$

Therefore, since the lim sup is at most 0 and the terms in $\int |f - f_n|$ are non-negative, both the lim sup and the lim of this sequence must be exactly 0. So, $|\int f - \int f_n| \leq \int |f - f_n| \rightarrow 0$ so $|\int f - \int f_n| \rightarrow 0$ so $\int f_n$ approaches $\int f$ so the limit and integral interchange is justified.

□

Now, if $f_n(x)$ is defined as n for $0 < x < 1/n$ then the integral of this will be 1, but at all points between 0 and 1 f_n will eventually go to 0, so the integral of the limit is not the limit of the integral. It turns out that in this case it turns out we cannot find a function g such that for all n we have $|\int_0^1 f_n| \leq |\int_0^1 g| < \infty$. This is a standard textbook counterexample.

2.3 Fubini's theorem

We will only prove a special case of Fubini's theorem.

Another result says that if I have $\iint f(x, y) dx dy$ on a bounded rectangle then I can swap the integrals provided f is continuous. Specifically, $\int_{y_0}^{y_1} \left(\int_{x_0}^{x_1} f(x, y) dx \right) dy = \int_{x_0}^{x_1} \left(\int_{y_0}^{y_1} f(x, y) dy \right) dx$

The claim above is called Fubini's theorem.

A definition: Let f be any bounded function on $[a, b]$, then $\int_a^b f(t) dt$ is the supremum over all partitions of $[a, b]$ of the integrals of sums of rectangles that underestimate the function at each partition. This is called the lower integral. We similarly define the upper integral as $\overline{\int_a^b f(t) dt}$. f is said to be Riemann integrable if these integrals are equal.

Proof. We will use grid partitions, the X component will be $X_1, X_2, \dots, X_n$ and the Y component will be $Y_1, Y_2, \dots, Y_m$. We will define the lower integral as

$$\begin{aligned} & \sum_{i,j} |X_i \times Y_j| \inf_{x \in X_i, y \in Y_j} f(x, y) \\ & \leq \sum_i |X_i| \inf_{x \in X_i, y \in Y_j} \sum_j |Y_j| \inf_{y \in Y_j} f(x, y) \\ & \leq \sum_i |X_i| \inf_{x \in X_i} \int_Y f(x, y) dy \end{aligned}$$

$$\begin{aligned}
&\leq \sum_i |X_i| \sup_{x \in X_i} \int_Y f(x, y) dy \\
&\leq \sum_i |X_i| \sup_{x \in X_i} \sum_j |Y_j| \sup_{y \in Y_j} f(x, y) \\
&= \sum_{i,j} |X_i \times Y_j| \sup_{x \in X_i, y \in Y_j} f(x, y)
\end{aligned}$$

if f is continuous then I can make the second and fifth line differ by an arbitrarily small amount as I refine the partition. I could also swap the roles of X and Y in the proof above. and the middle 2 lines would correspond to the different orders of doing the integrals.

□

Definition. Let g be a function, then we define the Riemann Stieljes integral of $\int f(x) dg(x)$ as instead of taking $x_i - x_{i-1}$ we take $g(x_i) - g(x_{i-1})$. When we integrate with respect to a probability measure in Level 6.3, we are essentially setting g to the cumulative distribution function and redefining the “length” for the Lebesgue theorems.

Proposition. If g is a cumulative distribution function and f is continuous then f is integrable on any bounded interval.

Proof. The same as the level 4 proof that continuous functions are integrable relied on the fact that showing if our intervals get small enough f will differ by at most ε on those intervals. We assumed without loss of generality that the integrals are from 0 to 1 and concluded from there, and the proof is exactly the same since the total change of g is never more than 1 as it is a cumulative probability distribution.

□

Note that for improper integrals, provided the integral of the absolute value is finite, then if we make a big enough rectangle the absolute value of the outside things will go to 0, and the limit will not depend on what sequence of rectangles we take. Therefore for improper integrals we will check for absolute convergence.

3 Intuition for complex integration

Also, in my exponentials and logarithms video, I briefly mentioned that if a function has a single-valued antiderivative then the integral along a path of that function does not depend on the path. By “integrate along a path”, I mean take the sum of the function times the distance you move by on the path (eg if you move from $1.07 + 3.12i$ to $1.08 + 3.14i$ you add $(0.01 + 0.02i)f(1.07 + 3.12i)$), then take the limit of these sums as these distances go to 0. Intuitively, similar to actual integration, taking the sum like this moves you along the antiderivative, which if it is single valued (unlike log, for example) will not allow you to possibly have path dependence.

4 Uniform continuity

Definition. To say A function is Uniformly continuous means that not only are we able to pick a δ for each x , but a δ that works for all x . An example of a function that is not uniformly continuous is $1/x$ on the open interval $(0,1)$, since if I pick a certain ε , then no matter how small I pick δ , I can go close enough to 0 that the $|x - a| < \delta$ implies $|f(a) - f(x)| < \varepsilon$ condition is not satisfied, so I cannot pick a δ that works for all x .

Theorem. Suppose that f is a function which is continuous on a closed bounded interval, then it is uniformly continuous

Proof. Since we are assuming our function is continuous on a closed bounded interval, the Bolzano Weierstrass theorem applies by boundedness, and the subsequence in question converges to a limit in the interval by closure of the interval.

We will see similarities to the level 4 proof that continuous functions are integrable.

Assume for a contradiction that our function is continuous but not uniformly continuous on our closed bounded interval. Then there is an $\varepsilon > 0$ such that if I pick $\delta = \frac{1}{n}$ there is some x_n, y_n with $|x_n - y_n| < \frac{1}{n}$ but $|f(x_n) - f(y_n)| > \varepsilon$. We know that x_n and y_n both have convergent subsequences converging to places on our closed interval, and in fact they converge to the same place since the difference between x_n and y_n gets arbitrarily small. If the subsequences in question are x_{n_i} and y_{n_i} and their limits are x , then $f(x_{n_i}) \rightarrow f(x)$ and $f(y_{n_i}) \rightarrow f(x)$ because continuous functions have the property that whenever the inputs are close the outputs are close, so intuitively you can pass limits through continuous functions like this. But $f(x_{n_i})$ and $f(y_{n_i})$ somehow converge to the same limit but are always $> \varepsilon$, which is a contradiction.

□

5 Cardinality

Now, I will talk about cardinality. This is really fun stuff, but I will not go through all the fun results here, only the results we need. The motivation of this is to show that there are, in a precise sense, “more” real numbers than natural numbers, so I can show that you cannot take a sum over all the real numbers of positive numbers and still get a finite number.

We consider two sets to be the same size if there is a one to one correspondence between their elements. This sounds obvious, but it shows that there are equally as many even positive integers as positive integers, because we can map 1—2 and 2—4 and 3—6 and so on, giving us a one to one correspondence.

Theorem. We cannot find a one to one correspondence between real numbers on any interval and positive integers.

The proof of this theorem is very famous in “popular math” and the slogan is “there are different sizes of infinity”.

Proof. We will do this for the interval (0,1) then note that we can get to any interval by scaling, or the entire reals by doing $-\cot(\pi x)$ for example. The property of there being a correspondence to positive integers is called being countable. But, I think listable is a better term.

So, suppose we do, in fact, have a list of all the real numbers between 0 and 1.

1. 0.3141592653589793...
2. 0.2718281828459045...
3. 0.1414213562373095...
4. 0.3498579345858968...
5. 0.9988237478199283...

Now construct a new number where the n 'th decimal digit after the point is not the n th digit of the n th number, and such that we do not have infinite trailing 9's or 0's (to ensure the number has a unique decimal representation. For the numbers in our list we pick the one with trailing 0's and not trailing 9's whenever we have to make this choice). For example, adding 1 to each red we could have 0.48293... But this is clearly not on the list – It differs from everything on the list by at least one digit. Contradiction. Essentially, this makes it so if you think you found a way to list them, I can prove you are wrong.

□

We want to prove that you can't add an unlistable amount of real numbers greater than 0 and get a finite result.

Now if we had a sum over the real numbers of positive numbers x , split them into subsets:

$$x \geq 1, \frac{1}{2} \leq x < 1, \frac{1}{3} \leq x < \frac{1}{2}, \frac{1}{4} \leq x < \frac{1}{3}.$$

Then one of these subsets has to have infinitely many elements. Why? If they all had finitely many elements then they would form a subset of this infinite table:

	1	2	3	4	5	6	7	8	9	...
Set 1										
Set 2										
Set 3										
Set 4										
Set 5										
Set 6										
Set 7										
Set 8										
...										

So I could then go along the table in a zigzag pattern like this in the order shown below, adding to my list whenever the elements are in the table, and not adding them otherwise.

	1	2	3	4	5	6	7	8	9	...
Set 1	1	2	6	7	15	...				
Set 2	3	5	8	14						
Set 3	4	9	13							
Set 4	10	12								
Set 5	11									
Set 6										
Set 7										
Set 8										
...										

All cells get reached eventually, so this would imply we can list the cells. But we are assuming we are adding one term for each real number, and we cannot list the real numbers. Therefore, one of the sets

$$x \geq 1, \frac{1}{2} \leq x < 1, \frac{1}{3} \leq x < \frac{1}{2}, \frac{1}{4} \leq x < \frac{1}{3}.$$

has infinitely many elements, so the sum is infinite.

6 Calculus results

6.1 Big O and little o notation

Definition. A function $g(x)$ is said to be $O(f(x))$ if when x is sufficiently close to x_0 , $|g(x)|$ is bounded by $M|f(x)|$ where M is some fixed positive constant. If $f(x)$ is not 0 in the vicinity of x_0 we can write that $|\frac{g(x)}{f(x)}| < M$ when x is sufficiently close to x_0 . We can also say, and this is often done in computer science when analyzing how long an algorithm will take, that a function is a function $g(x)$ is $O(f(x))$ as x goes to infinite if when x is large enough, $|g(x)|$ is bounded by $M|f(x)|$. The value of x_0 in a specific situation is often implied by context. Sometimes people write $g(x) = O(f(x))$

Definition. Little o notation is like big O notation except if $g(x)$ is $o(f(x))$ at x_0 it means that $g(x)$ gets much smaller than $f(x)$ when x is sufficiently close to x_0 . Precisely, it means that M can be made arbitrarily small by making x sufficiently close to x_0 , or by making x sufficiently large in the case x_0 is infinity.

Note that if $f(x) = o(g(x))$, then we necessarily also have $f(x) = O(g(x))$.

Example. $x \neq O(x^2)$ as $x \rightarrow 0$ since x^2 is much smaller than x . In fact, $x^2 = o(x)$ as $x \rightarrow 0$.

$x = O(x^2) = o(x^2)$ as $x \rightarrow \infty$ but $x^2 \neq O(x)$ as $x \rightarrow \infty$.

$x = O(\sqrt{x})$ as $x \rightarrow 0$.

$\sin(2x) = O(x)$ as $x \rightarrow 0$, but $\sin(2x) \neq o(x)$ as $x \rightarrow 0$.

$x \neq O(x \sin(x))$ as $x \rightarrow \infty$, because although x does not seem to be much larger than $x \sin(x)$, the ratio $\frac{x}{x \sin(x)}$ is unbounded when x gets close to integer multiples of π .

Also, a principle here is the idea that a polynomial is dominated by its leading term. For example

$2x^3 + 4x + 12 = O(x^3)$ as $x \rightarrow \infty$ because clearly eventually x^3 exceeds $4x + 12$ so the whole thing is less than $3x^3$ so we can put $M = 3$. Also, big o notation can capture the idea that exponentials beat polynomials (by the way you can prove this as $\frac{a^x}{x^b}$ goes to infinity as x goes to infinity which you can see if you apply l'hospital's rule at least b times), as we can write that $P(x) = o(\exp(x))$ for any polynomial P . Also, $\log(x) = o(x^\varepsilon)$ for all positive ε and this can be proven the same way, or by thinking of it intuitively as a sort of inverse of the exponentials beat polynomials idea. Note $\exp(x)$ means e^x .

Another idea is that non-zero constants don't matter here since we're looking at the bigger picture with the growth rate of these functions.

Example. Since $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} = f'(x)$, $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} - f'(x) = 0$ so $\frac{f(x+h)-f(x)}{h} - f'(x) = o(1)$ as x goes to h by definition. Therefore, $f(x+h) - f(x) - hf'(x) = o(h)$ by multiplying by h on both sides, so we can write that $f(x+h) - f(x) = hf'(x) + o(h)$, which looks a lot like what I did when I justified differentiating infinite power series.

6.2 Taylor's theorem

Definition. A Taylor polynomial of degree n is a Taylor series for a function that is n times differentiable but only up to the term in x^n . We write this as $T_n(f)(x)$. This is a polynomial which gives an approximation of f near some point x_0 , with error equal to $o((x - x_0)^n)$.

Claim. If f is n times differentiable at x_0 then $f(x) - T_n(f)(x) = o((x - x_0)^n)$ as $x \rightarrow x_0$ if x_0 is the point we are doing a Taylor series around.

Claim. If f is $n+1$ times continuously differentiable on the interval $[x_0, x]$ then $f(x) - T_n(f)(x) = O((x - x_0)^{n+1})$ as $x \rightarrow x_0$ if x_0 is the point we are doing a Taylor series around.

Claim. If the conditions for claim 2 hold, then there exists some $t \in (x_0, x)$ such that $f(x) - T_n(f)(x) = \frac{(x-x_0)^{n+1}}{(n+1)!} f^{(n+1)}(t)$.

We will prove this.

Lemma. (Extreme value theorem) Any function continuous on a finite closed interval is bounded and actually attains its minimum and maximum.

Proof. Note: It is very important that the interval is closed. On the open interval $(0, 1)$ we can define $1/x$ which is continuous on that interval, but not on the closed interval $[0, 1]$ because the closed interval contains $x = 0$ but the open interval does not. But there's not much you can do to make a continuous unbounded function defined on a finite closed interval – Try it!

We will use the fact that every bounded sequence has a convergent subsequence since that was proven in level 4.

Suppose for a contradiction f is unbounded. Then for all n , we can choose x_n in $[a, b]$ with $|f(x_n)| > n$. Now pick a convergent subsequence x_{n_k} of the sequence formed by each x_n , and call its limit L , which is in $[a, b]$ since $[a, b]$ is closed so limits of sequences in $[a, b]$ are in $[a, b]$. By continuity, $f(x_{n_k}) \rightarrow f(L)$, but since $f(x_{n_k})$ is unbounded, it follows that this is a contradiction, since $f(L)$ cannot be defined.

Important note: Bolzano weierstrass holds for sequences of points in k dimensions, provided the bounded set the sequence is in is closed as mentioned above – there is a subsequence whose x coordinate converges and a subsequence of that whose y coordinate converges etc, and thus the above theorem holds for functions of more than one variable, as well as the theorem about uniform continuity and about riemann integrals for continuous functions, as the definition of continuity is analogous. This is important since we use these theorems in the context of functions of multiple variables.

We will now prove that the maximum is attained. By boundedness there is a least upper bound M . Therefore there is a sequence of increasing points $M_1, M_2, \dots$ converging to M that are values of the function. For each of these points pick an input value t_k of f such that $f(t_k) = M_k$. Now t_k has a convergent subsequence converging to t and by continuity $f(t) = M$, meaning the maximum value of a function on a closed interval is attained. It is by closedness that we can guarantee t is in our interval.

□

Lemma. (Mean value theorem)

If we have a continuous function on $[a,b]$ differentiable on the open interval (a,b) then at some point in (a,b) the derivative is equal to its mean value.

This lemma assumes the values are real, so everything proven here applies to real numbers, but not general complex numbers.

Proof. First of all, a picture (Figure 1, from wikipedia):

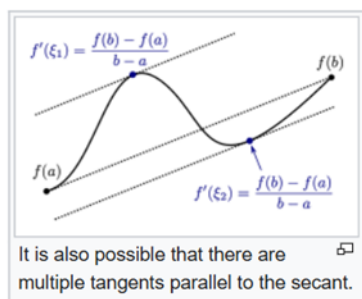
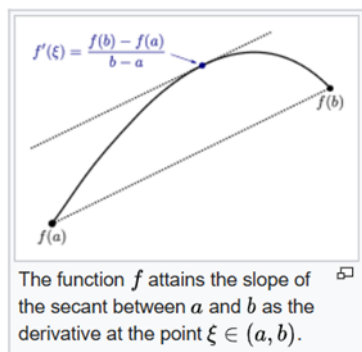


Figure 1

You can see that it's kind of obvious that the function's derivative attains its mean value, since the function is differentiable everywhere by assumption so there are no spikes. This is essentially what the theorem says. However, it is not completely obvious as derivatives can be discontinuous.

To prove this, we want to subtract the line from $(a, f(a))$ to $(b, f(b))$ from y . This line has the slope we want so we want to prove that our new function, which is 0 at a and b , attains a derivative of 0 between a and b . This new function is either constant, in which case the theorem is obvious, or it goes above 0 at some point (If not prove the theorem for minus the function). Since it is continuous on a closed, bounded interval, it is bounded, and has either a maximum or a minimum. At such a point, the derivative exists by our hypothesis, and it can't possibly be something other than 0. Why? Suppose the derivative were positive at a maximum point c , then by definition of limits, $\frac{f(x) - f(c)}{x - c}$ would be positive if we make x close enough to c with $x > c$, implying $f(x) > f(c)$ which is a contradiction.

In particular, we have Rolle's theorem, which says that if $f(a) = f(b) = 0$ and f is differentiable on (a, b) then there is a point $a < x < b$ such that $f'(x) = 0$

□

Lemma. Let f be continuous on $[a, b]$ and $n-1$ times continuously differentiable, but also n times differentiable on the open interval (a, b) , and suppose also that

$$f(a) = f'(a) = f''(a) = \dots = f^{(n-1)}(a) = f(b) = 0$$

then there is some x in (a, b) such that $f^{(n)}(x) = 0$.

Proof. Induct on n . The $n = 0$ base case is just Rolle's theorem. Suppose we have $k < n$ and $x_k \in (a, b)$ such that $f^{(k)}(x_k) = 0$. Since $f^{(k)}(a) = 0$, we can find $x_{k+1} \in (a, x_k)$ such that $f^{(k+1)}(x_{k+1}) = 0$ by Rolle's theorem. So the result follows by induction. □

Example. The function $x^3 - x^2$ has its first derivative and value equal to 0 when $x = 0$, and its value equal to 0 at $x = 1$, so between 0 and 1 it must have a point of inflection. Indeed it does when $x = \frac{1}{3}$.

Proposition. If $f(x)$ is n times differentiable at $x = x_0$ then

$$f(x) - f(x_0) - (x - x_0)f'(x_0) - \frac{(x - x_0)^2}{2!}f''(x_0) - \dots - \frac{(x - x_0)^n}{n!}f^{(n)}(x_0)$$

is $o((x - x_0)^n)$ as $x \rightarrow x_0$.

Proof. The first n derivatives of the above expression are all 0 when evaluated at x_0 because for k between 0 and n inclusive the $\frac{(x - x_0)^k}{k!}f^{(k)}(x_0)$ term has k 'th derivative equal to $f^{(k)}(x_0)$ and all its other derivatives are 0 at x_0 by repeated application of the power rule so therefore it cancels the k 'th derivative of the $f(x)$ term to make it equal to 0.

So, by the definition of o , we need to show that $\frac{f(x) - f(x_0) - \dots - \frac{(x - x_0)^n}{n!}f^{(n)}(x_0)}{(x - x_0)^n} \rightarrow 0$ as $x \rightarrow x_0$. This is fine as $(x - x_0)^n$ is not 0 when you make it arbitrarily small so it can go in the denominator.

Now, if we can show that any function whose first n derivatives are all 0 at x_0 is $o((x - x_0)^n)$, we will be done. It is clear that a function whose value and first derivative is 0 at x_0 is $o(x - x_0)$ since $\frac{g(x)}{x - x_0} = \frac{g(x) - g(x_0)}{x - x_0} \rightarrow g'(x_0) = 0$, so this theorem is true if $n=1$. But now suppose that we know that this theorem is true for $n=k$, then we will prove it is also true for $n=k+1$, therefore proving this by induction. Let $g(x)$ be such that at x_0 its value and first $k+1$ derivatives are all 0. By the induction hypothesis, $g'(x)$ has its first k derivatives and value equal to 0 at x_0 so $g'(x_0 + h) = h^k \varepsilon(h)$ with $\varepsilon(h) \rightarrow 0$ as $h \rightarrow 0$, by the definition of little o notation and the induction hypothesis. By the mean value theorem, there exists some ϑ between 0 and 1 such that

$g'(x_0 + \vartheta h) = \frac{g(x_0 + h) - g(x_0)}{h}$, or equivalently $hg'(x_0 + \vartheta h) = g(x_0 + h) - g(x_0)$, meaning that by the induction hypothesis $g(x_0 + h) = h(h^k \varepsilon(\vartheta h)) = h^{k+1} \varepsilon(\vartheta h) = o(h^{k+1})$. So induction done and proof complete. □

Remark:

Just because each remainder is small does not mean that the Taylor series converges, since convergence at some x requires that there, the remainder is uniformly small, but it could be that the interval where the remainder is small shrinks and is zero in the limit. There are functions such that the Taylor series does not converge in any interval despite the functions being infinitely differentiable, like $\exp(-x^{-2})$ around $x=0$. However, it turns out that if the function has a complex derivative the Taylor series always converges in some interval, see complex analysis in a later level. I'm not proving it because we don't need to use it for this purpose, but it trivializes all of these results.

Lemma. Let f be $n+1$ times differentiable on $[a, b]$ with its $(n+1)$ 'th derivative continuous. Then we have that

$$f(b) = f(a) + (b - a)f'(a) + \frac{(b - a)^2}{2!}f^{(2)}(a) + \dots + \frac{(b - a)^n}{n!}f^{(n)}(a) + \int_a^b \frac{(b - t)^n}{n!}f^{(n+1)}(t) dt$$

Proof. We will do this by induction on n . When $n=0$ the theorem says

$$f(b) = f(a) + \int_a^b f'(t) dt$$

Which is known to be true. Now we will do integration by parts to get

$$\int_a^b \frac{(b - t)^n}{n!}f^{(n+1)}(t) dt = \left[\frac{-(b - t)^{n+1}}{(n + 1)!}f^{(n+1)}(t) \right]_a^b + \int_a^b \frac{(b - t)^{n+1}}{(n + 1)!}f^{(n+2)}(t) dt$$

$$= \frac{(b-a)^{n+1}}{(n+1)!} f^{(n+1)}(a) + \int_a^b \frac{(b-t)^{n+1}}{(n+1)!} f^{(n+2)}(t) dt$$

So the lemma follows by induction. □

Proposition. If $f(x)$ is $n+1$ times continuously differentiable in an open interval containing x_0 then

$$f(x) - f(x_0) - (x-x_0)f'(x_0) - \frac{(x-x_0)^2}{2!}f''(x_0) - \dots - \frac{(x-x_0)^n}{n!}f^{(n)}(x_0) - \dots$$

is $O((x-x_0)^{n+1})$ as $x \rightarrow x_0$.

Proof. The lemma above immediately implies the result since $f^{(n+1)}$ is continuous on the closed interval $[a, b]$ and thus bounded there, say it is bounded in absolute value by M . Then the integral is at most $\int_a^b \frac{(b-t)^n}{n!} M dt = M \frac{(b-a)^{n+1}}{n+1!}$ which is $O((x-a)^{n+1})$ as required. □

Proposition. $f(x) - f(x_0) - (x-x_0)f'(x_0) - \frac{(x-x_0)^2}{2!}f''(x_0) - \dots - \frac{(x-x_0)^n}{n!}f^{(n)}(x_0) = \frac{(x-x_0)^{n+1}}{(n+1)!}f^{(n+1)}(t)$ for some $t(x)$ between x_0 and x , provided the first $n+1$ derivatives of f exist and are continuous from x_0 to x . (In fact we don't need the $n+1$ 'th derivative to be continuous, but merely for it to exist on the open interval (x_0, x) as that is enough to apply Rolle's theorem).

Proof. The $n=0$ case is just the mean value theorem.

Consider $g(x) := f(x) - f(x_0) - (x-x_0)f'(x_0) - \frac{(x-x_0)^2}{2!}f''(x_0) - \dots - \frac{(x-x_0)^n}{n!}f^{(n)}(x_0)$. This has its value and first n derivatives at x_0 equal to 0 by the same reasoning as we gave in one of the earlier proofs

Set $\frac{g(x)}{(x-x_0)^{n+1}} = C$ when x is our constant. Then $g(x) - C(x-x_0)^{n+1}$ as a function of x (so x is no longer constant) is 0 at x from how we defined C and has its first n derivatives and value equal to 0 at x_0 meaning it satisfies the conditions for generalized Rolle's theorem. Therefore, there exists a point y between x_0 and x such that the $n+1$ 'th derivative of $g(y) - C(y-x_0)^{n+1}$ is 0. But the derivative of the second term is just $C(n+1)!$ by the power rule, thus the $n+1$ 'th derivative of g at y is also $C(n+1)!$, thus $C = \frac{g^{(n+1)}(y)}{(n+1)!}$. Therefore, since we have that $g(x) - C(x-x_0)^{n+1}$ is 0 at x from earlier, we thus have that

$$g(x) = C(x-x_0)^{n+1} = \frac{g^{(n+1)}(y)}{(n+1)!}(x-x_0)^{n+1}$$

Which is exactly what we wanted to prove. □

Example. There exists some point between 0 and 1 such that $\frac{e^x}{2} = e-2$, because if $x=1$ and $x_0=0$, then $e^x - x - 1 = \frac{x^2}{2}e^y$ for some y between 0 and 1, but $x=1$ so this simplifies to $\frac{e^y}{2} = e-2$.

6.3 L'hospital's rule

Lemma. (Cauchy mean value theorem) Let f, g be functions taking real values continuous on $[a, b]$ and differentiable (a, b) . Then there exists a c in (a, b) such that $g'(c)(f(b) - f(a)) = f'(c)(g(b) - g(a))$

Proof. Try the function $h(x) = [g(x) - g(a)][f(b) - f(a)] - [g(b) - g(a)][f(x) - f(a)]$.

Now $h'(x) = g'(x)(f(b) - f(a)) - f'(x)(g(b) - g(a))$

This is 0 at some point because $h(a) = h(b)$ and Rolle's theorem is true.

Proof of L'hospital's rule

Suppose that the following hold:

1. $f(a) = g(a) = 0$
2. $g(x)$ is not 0 if x is close to c
3. $\lim_{x \rightarrow c} \frac{f'(x)}{g'(x)}$ exists and is equal to l .
4. f and g are differentiable near a

By the Cauchy mean value theorem $g'(c)(f(b)) = f'(c)(g(b))$ for some c between a and b , for any $b > a$. Rearranging gives that

$$\frac{f(b)}{g(b)} = \frac{f'(c)}{g'(c)}$$

I emphasise that c depends on b but for each fixed b it exists and is between a and b . Therefore, as b approaches a , so does c .

Thus, $\frac{f(b)}{g(b)} \rightarrow l$ as $b \rightarrow a$.

7 Uniform convergence

We will define uniform convergence.

We know about how pointwise convergence works – a sequence of functions converges to a function if it converges at every value. We can say “For each point, there is an $\varepsilon > 0$ such that for all x , our sequence of functions $f(x)$ is eventually, after some n , within ε of the limit function”. This is different from uniform convergence in the sense that uniform convergence requires n not to depend on x . We need the function to eventually get arbitrarily close at all points at once, not just at each point. An example of a sequence of functions that converges pointwise but fails this is the following:

$$f_n(x) = \begin{cases} 0 < x < \frac{1}{n} : 1 - nx \\ \text{Otherwise} : 0 \end{cases}$$

This looks like the functions in these images:

Figure 2 shows examples of this for $n=1, 2, 10$.

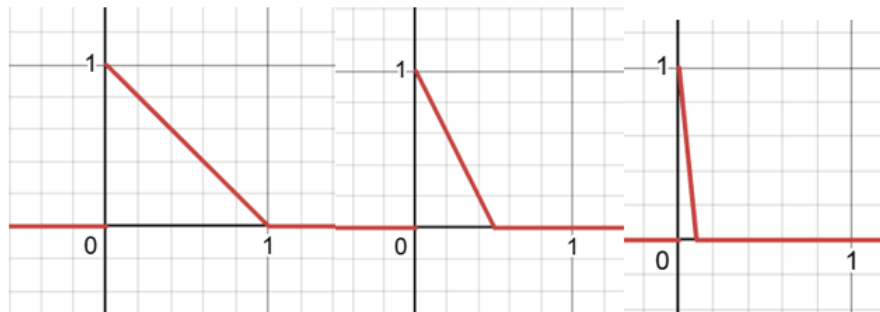


Figure 2

The point is: For any point you choose, say k , after $> \frac{1}{k}$ steps, we will get to 0. So we converge pointwise to 0. It seems like we can pick $x=0$ and it will converge to 1 but we defined $f(0)$ to be 0. We do not converge uniformly to 0 as we never satisfy that all points get arbitrarily close to 0.

It is useful because, for example, if we want to swap a limit with an integral we know that the total error will eventually get small, rather than just the error at each point.